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Aim :- Implement and analyze various probability distributions in python, including Binomial, Poisson, and Normal distributions.

1. What is difference between Binomial, Poisson, and Normal distributions?

Binomial, Poisson, and Normal distributions are all probability distributions but used in different situations.

Binomial distribution is used when you have a fixed number of trials, each with two outcomes like success/failure, and the probability of success stays the same. Example: number of heads in 10 coin tosses.

Poisson distribution is used to count how many times an event happens in a fixed interval of time or space, when events occur randomly and independently. Example: number of calls received in an hour.

Normal distribution is a continuous distribution that is symmetric and bell-shaped, used for measurements like height, weight, or test scores. It is defined by mean and standard deviation.

2. What is the real world significance of the Poisson distribution?

The Poisson distribution is important in real life because it helps model the number of times an event occurs in a fixed interval of time or space when events happen randomly and independently.

Its significance lies in practical applications such as:

predicting the number of customer arrivals in a shop or calls in a call center
estimating the number of accidents at a location
modeling defects in manufacturing processes
analyzing arrival of emails, messages, or network traffic

It helps organizations plan resources, improve efficiency, and make decisions under uncertainty.

3. What is the Central Limit Theorem and how does it relate to Normal distribution?

The Central Limit Theorem states that when you take a large number of independent random samples from any distribution, the distribution of their sample means tends to become approximately Normal, regardless of the original distribution.

Its relation to the Normal distribution is that it explains why the Normal distribution appears so often in real life. Even if the original data is not normally distributed, the averages of large samples will follow a bell-shaped (Normal) distribution.

4. Why is Normal distribution widely used in machine learning and statistics?

The Normal distribution is widely used in machine learning and statistics because many real-world data sets naturally follow a bell-shaped pattern, such as heights, test scores, and measurement errors.

One key reason is mathematical simplicity. The Normal distribution is easy to work with and has well-defined properties using just mean and standard deviation, which makes calculations and modeling more efficient.

Another reason is the Central Limit Theorem, which states that averages of large samples tend to follow a Normal distribution even if the original data is not normal. This makes it very useful for making inferences and predictions.

It is also commonly used to model noise and errors in data, which helps improve the performance of many machine learning algorithms

5. What is Skewness in distributions and which distributions exhibit it?

Skewness is a measure of how much a probability distribution is asymmetric around its mean.

If the distribution is symmetric (like the Normal distribution), skewness is zero. If it is positively skewed (right-skewed), the tail is longer on the right side (mean > median). If it is negatively skewed (left-skewed), the tail is longer on the left side (mean < median).

Distributions that exhibit skewness include:

Poisson distribution (usually positively skewed, especially when the average is small)
Exponential distribution (positively skewed)
Binomial distribution (can be skewed depending on probability; symmetric only when probability = 0.5)

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Set up figure
fig, axs = plt.subplots(3, 3, figsize=(15, 12))
fig.suptitle("Analysis of Binomial, Poisson, and Normal Distributions")
```

```

# ----- BINOMIAL DISTRIBUTION -----
n, p = 10, 0.5
x_binom = np.arange(0, n+1)
pmf_binom = binom.pmf(x_binom, n, p)
cdf_binom = binom.cdf(x_binom, n, p)
samples_binom = binom.rvs(n, p, size=1000)

# PMF
axs[0, 0].bar(x_binom, pmf_binom)
axs[0, 0].set_title("Binomial PMF (n=10, p=0.5)")

# CDF
axs[0, 1].step(x_binom, cdf_binom, where='mid')
axs[0, 1].set_title("Binomial CDF")

# Histogram
axs[0, 2].hist(samples_binom, bins=np.arange(n+2)-0.5, density=True)
axs[0, 2].set_title("Binomial Samples Histogram")

# ----- POISSON DISTRIBUTION -----
mu = 4
x_poisson = np.arange(0, 15)
pmf_poisson = poisson.pmf(x_poisson, mu)
cdf_poisson = poisson.cdf(x_poisson, mu)
samples_poisson = poisson.rvs(mu, size=1000)

# PMF
axs[1, 0].bar(x_poisson, pmf_poisson)
axs[1, 0].set_title("Poisson PMF ( $\mu=4$ )")

# CDF
axs[1, 1].step(x_poisson, cdf_poisson, where='mid')
axs[1, 1].set_title("Poisson CDF")

# Histogram
axs[1, 2].hist(samples_poisson, bins=np.arange(16)-0.5, density=True)
axs[1, 2].set_title("Poisson Samples Histogram")

# ----- NORMAL DISTRIBUTION -----
mu_norm, sigma = 0, 1
x_norm = np.linspace(-4, 4, 1000)
pdf_norm = norm.pdf(x_norm, mu_norm, sigma)
cdf_norm = norm.cdf(x_norm, mu_norm, sigma)
samples_norm = norm.rvs(mu_norm, sigma, size=1000)

# PDF
axs[2, 0].plot(x_norm, pdf_norm)
axs[2, 0].set_title("Normal PDF ( $\mu=0$ ,  $\sigma=1$ )")

# CDF
axs[2, 1].plot(x_norm, cdf_norm)
axs[2, 1].set_title("Normal CDF")

# Histogram
axs[2, 2].hist(samples_norm, bins=30, density=True)
axs[2, 2].set_title("Normal Samples Histogram")

# Layout
for row in axs:
    for ax in row:
        ax.grid(True)

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()

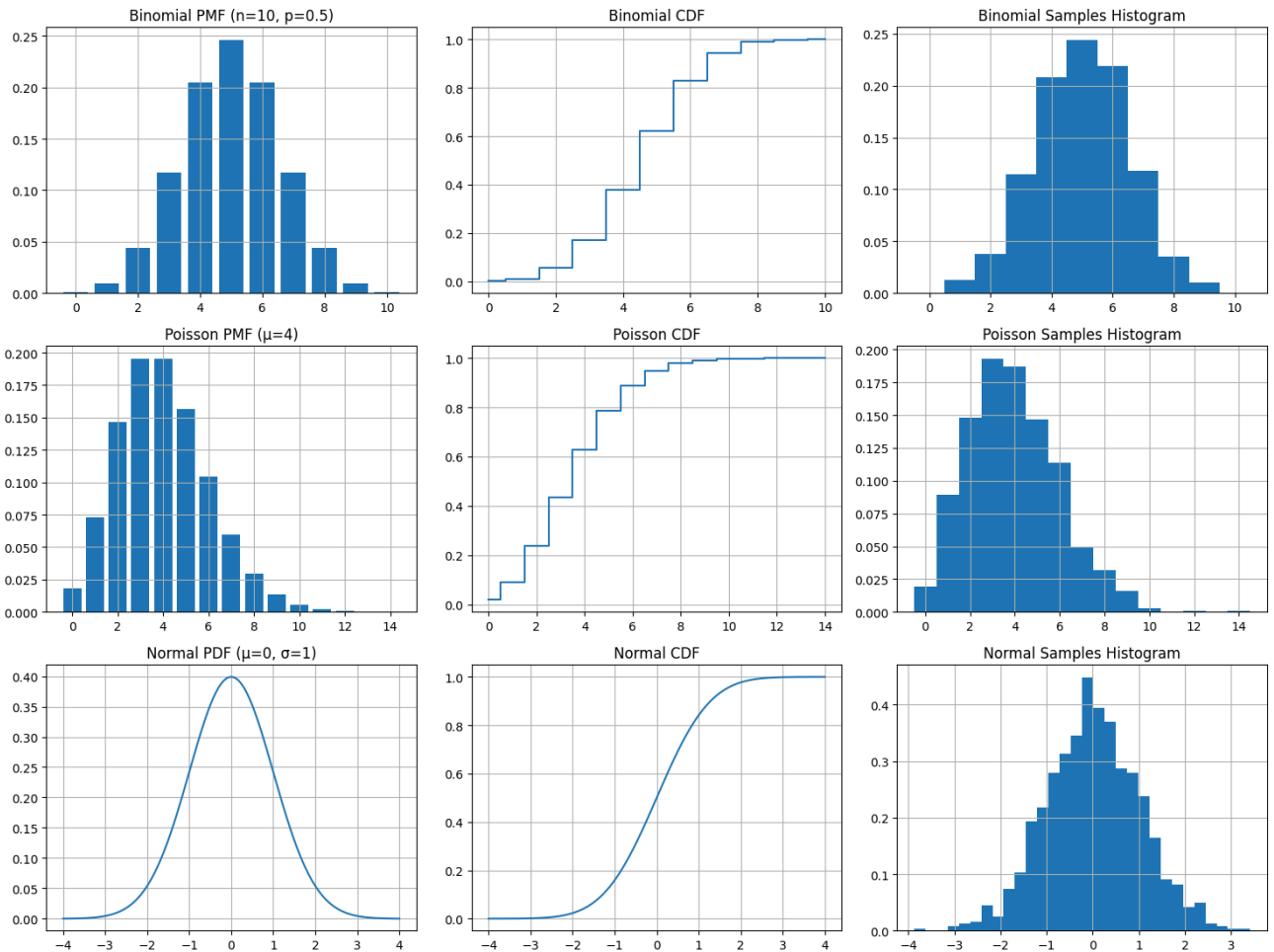
# ----- STATISTICS -----
print("Binomial Distribution:")
print("Mean =", np.mean(samples_binom), "Variance =", np.var(samples_binom))

print("\nPoisson Distribution:")
print("Mean =", np.mean(samples_poisson), "Variance =", np.var(samples_poisson))

print("\nNormal Distribution:")
print("Mean =", np.mean(samples_norm), "Standard Deviation =", np.std(samples_norm))

```

Analysis of Binomial, Poisson, and Normal Distributions



Binomial Distribution:

Mean = 4.996 Variance = 2.383984

Poisson Distribution:

Mean = 3.937 Variance = 4.123031

Normal Distribution:

Mean = 0.005394494364538035 Standard Deviation = 1.0209289938614357

Conclusion:

In this experiment, we implemented and analyzed Binomial, Poisson, and Normal distributions using Python. Each distribution was observed to model different types of real-world data effectively. The Binomial distribution was suitable for situations with fixed trials and binary outcomes, the Poisson distribution helped model random event counts over a fixed interval, and the Normal distribution represented continuous data with a symmetric pattern.

Through visualization and computation, we understood the behavior, shape, and key parameters of each distribution. We also saw how changes in parameters affect their spread and skewness. Overall, this experiment highlighted the importance of selecting the appropriate distribution for accurate data modeling and analysis in statistics and machine learning.

